Python Workshop
Final Project
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Introduction:

The study, "Neural pattern similarity predicts long-term fear memory" by Renée M. Visser et al., investigates the relationship between neural patterns during fear learning and the long-term expression of fear memory using fMRI. The researchers found that trial-by-trial neural similarity could effectively predict the retention of learned fear across various stimuli. This study highlights the significance of specific brain regions in understanding fear memory formation and retention.

In our project, in order to replicate the original research conducted by Visser et al. (2013), we performed various statistical analyses following the preprocessing of fMRI data.
This approach allows us to validate the key findings of the original study regarding the involvement of specific brain regions in fear memory processing.

Methods:

preprocess:

Preprocessing of fMRI data is essential to remove artifacts and noise that can obscure the true neural signals, ensuring that the data accurately reflects brain activity related to cognitive tasks.

Therefore we performed:

- Slice time correction
- Motion correction
- Spatial smoothing
- Structural to Functional Registration
- Structural to MNI Registration
- Functional Normalization to MNI
- High-pass Filtering

It is worth noting that since we don't have FSL software or other fMRI data preprocessing tools, along with the aim of this project that is to improve our python skills, some of the corrections that have been implemented are not 100% correct. This may affect our statistical analysis results.

Mask & Data:

After preprocessing the data, we extracted Bold values from each ROI to a dataframe.

We used Harvard-Oxford Cortical Atlas and Harvard-Oxford Sub-Cortical Atlas for masking.

Statistical analysis:

We performed 2 main analyses, Pearson correlation and ANOVA.

1. Similarity matrix
2. Within stimulus correlation
3. Between stimulus correlation
4. Anova

Results & Discussion:

1. **Similarity Matrix (Learning Phase)**

   In the learning phase, we created a similarity matrix to examine how neural responses to each stimulus evolved over time. Since each of the six stimuli was presented seven times, we had a total of 42 responses per subject, allowing us to measure how similar the brain's activation patterns were across trials. By computing Pearson correlation matrices, we assessed whether certain stimuli evoked consistent neural patterns, indicating stronger encoding and categorization. This visualization helped us determine whether participants grouped or distinguished between stimuli based on their learned associations. At the group level, we averaged similarity matrices across all participants to identify broader trends in fear learning and memory consolidation. These results provide insight into whether fear-conditioned stimuli (CSneg) exhibit more stable neural representations compared to neutral (CSneut) or minimally reinforced (CSmin) stimuli, which is critical for understanding the neural mechanisms underlying learning and categorization. (figure 1)

   2. Within-Stimulus Comparison

   In this analysis, we calculated the neural similarity between repeated presentations of the same stimulus over time to assess the stability of its neural representation. First, we sorted the trials by stimulus type (CSneg, CSneut, CSmin) and then computed the Pearson correlation between consecutive trials of the same stimulus. After averaging the results for each participant, we computed the group-level mean and visualized changes across trials. We expected that reinforced stimuli (CSneg) would show higher similarity across repetitions, as the brain may process them more consistently due to their association with an electric shock. Conversely, non-reinforced stimuli (CSmin) were expected to show lower stability over time, as they lack strong reinforcement that encourages a consistent neural representation. The results suggest that high within-stimulus similarity indicates stable learning and reinforced neural representations, whereas low similarity may suggest a weaker or less meaningful neural representation. (figure 2)

   3. Between-Stimulus Comparison

   In this analysis, we calculated the neural similarity between different stimuli that belonged to the same learning category (e.g., a face and a house that were paired with an electric shock) while ensuring that only trials presented in the same order were compared. The goal was to examine whether the brain encodes stimuli based on their learned meaning rather than just their visual features. We expected that reinforced stimuli (CSneg) would exhibit higher neural similarity compared to neutral or non-reinforced stimuli (CSneut, CSmin), as the brain might group them into a common "threat" category. High neural similarity in CSneg would indicate that the brain generalizes across stimuli that predict similar outcomes, reflecting higher-order fear learning. In contrast, low similarity among neutral or non-reinforced stimuli would suggest that the brain does not associate them strongly, indicating weaker associative learning. (figure 3)

4. ANOVA (Analysis of Variance)

Lastly, we used ANOVA to test whether there were significant differences in responses across the 6 different stimuli. The goal of this test was to determine if there was enough evidence to suggest that the mean responses (like the number of licks or response times) were significantly different between the stimuli. If we found a significant result, we could conclude that the stimuli did not produce the same responses, which would point to the participants differentiating between them. If the ANOVA showed significant differences, we could then use post-hoc tests to pinpoint exactly which stimuli were different from each other. This helped us understand if there was a clear distinction in how participants responded to each stimulus (figure 4).

We are not analyzing our results and comparing them to results from articles, as the data we worked with is unaligned data. That is, not all measurements were taken at exactly the same times or under the same conditions, which prevents a direct comparison with studies based on aligned data. However, examples of our own results are provided, showing how response patterns varied across different stimuli throughout the experiment.

figure 1-

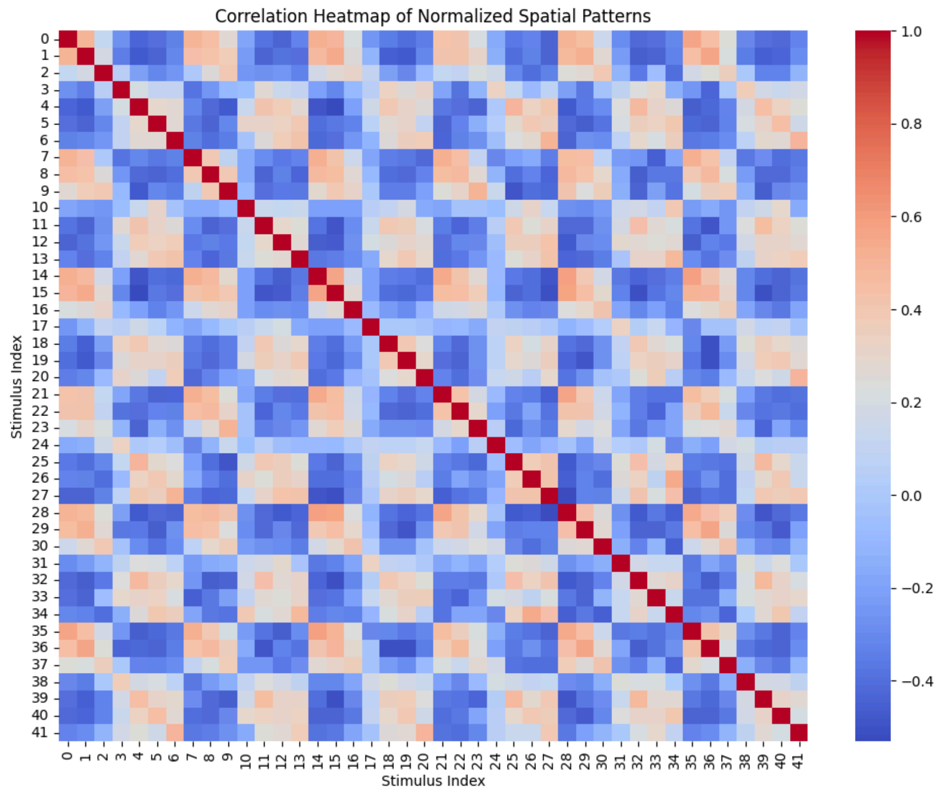

figure 2-

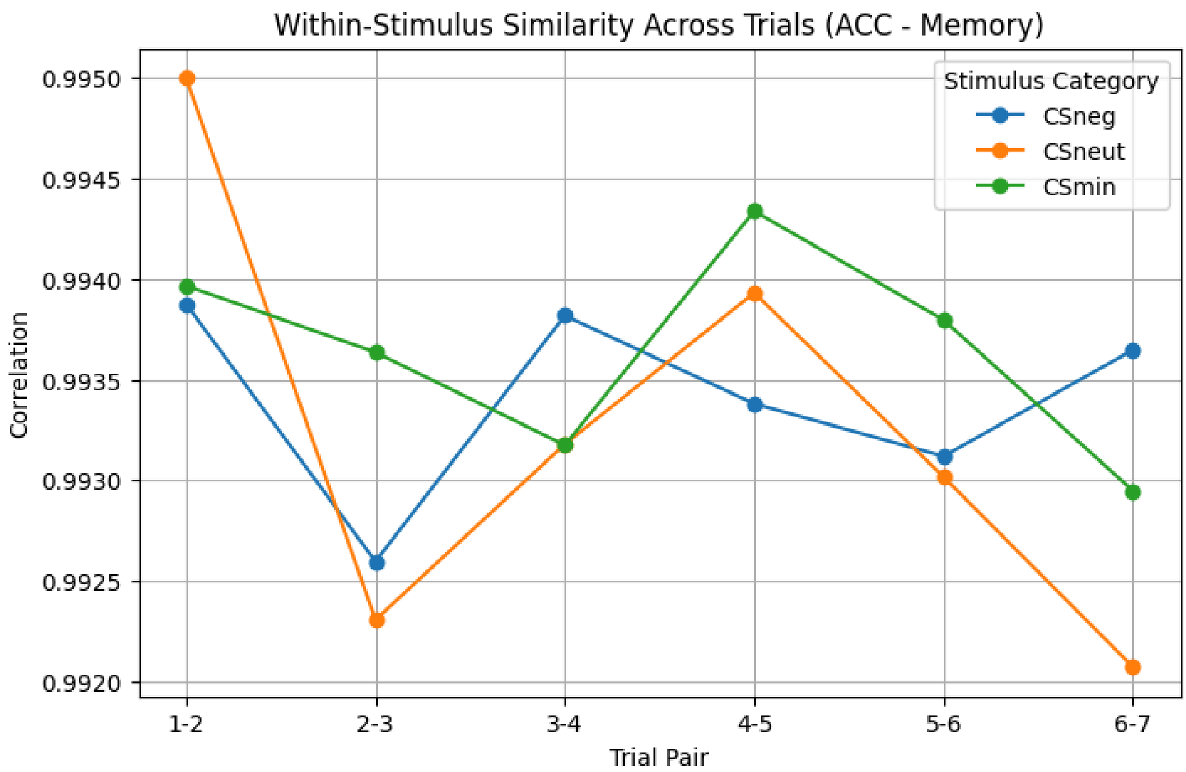

figure 3-

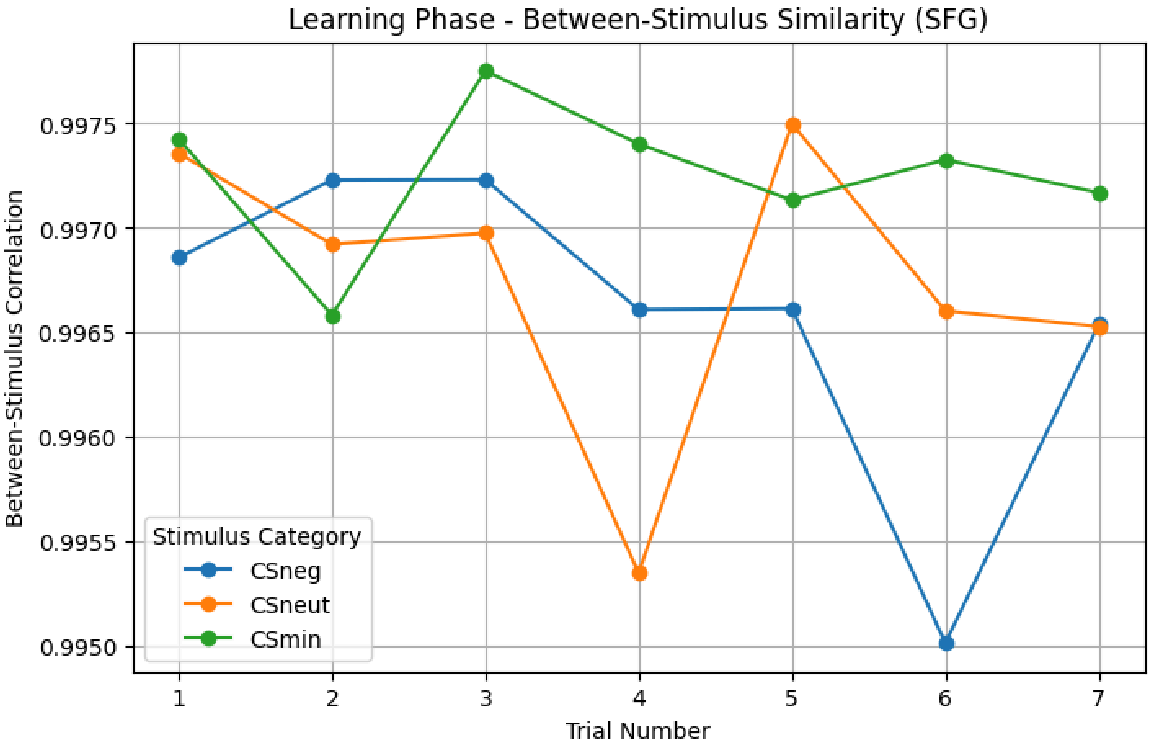

figure 4-